Missing citations in "A finite-entropy separation of microstates and nonmicrostates free entropy"

October 12, 2026

Summary of findings

Each row is a citation the preprint is missing: earlier work that already contains part of the preprint, which the preprint does not cite where that part appears. "Cited?" names the earlier work and says what, if anything, the preprint cites it for. The exception is a row flagged *overstated novelty*: there the preprint cites the work, but its abstract or framing presents the content as new. Terms are defined in Section 1; centrality is a best-effort judgment. Locations use the preprint's own numbering and printed pages.

ID	Relationship	Centrality	Technique or result	In the preprint	Cited?
F1	*same result*	central	Finite microstates entropy of a tuple with matrix models (here embeddable in $L(\mathbb{F}_2)$) plus small free semicircular noise	Lemma 4.2 (p. 9); models (p. 8)	Voiculescu 1994 [1]: cited for other content (definition of χ); Voiculescu 1990 [2], Jung 2002 [3], Jekel 2025 [4]: not cited; **attribution gap**
F2	*same result*	central	Microstates entropy bounded by the entropy of GUE-evolved microstate models minus the integrated free Fisher information, with the terminal entropies kept	Lemma 5.1, (13) (p. 10), proof (pp. 10–11); overview (p. 2)	Jekel–Pi 2023 [5]: credited in part (Prop. 6.1, pp. 2, 10; the elementary proof, p. 1); Shlyakhtenko–Tao 2020 [6]: credited elsewhere (through Jekel–Pi, p. 1); **overstated novelty**
F3	*same technique*	central	Fisher-deficit bound by lifting higher-order Stein kernels (here free cumulants) into semicircular chaos	Lemma 3.1, (5) for $r \geq 2$ (p. 5); Prop. 3.2 (pp. 6–7)	Diez 2023 [7], Fathi–Nelson 2016 [8], Cébron 2018 [9]: not cited; **attribution gap**
F4	*adapted*	central	Free semicirculars plus a tensor-independent coordinate and small free noise: finite χ, but matrix models that inherit the commutation lose entropy	§4 (p. 7); (12) (p. 9); (19), Lemma 6.1, proof of Theorem 1.1 (pp. 12–14)	Jekel 2025 [4]: not cited; **attribution gap**

ID	Relationship	Centrality	Technique or result	In the preprint	Cited?
F5	*adapted*	central	Unitary-net covering of matrix tuples with a common block structure	Lemma 6.1, Gaussian bound (p. 12; proof p. 13)	Voiculescu 1996 [10] (as expounded by Biane 2001 [11] and Mingo–Speicher 2017 [12]), Dykema 1996 [13]: not cited; Ge 1996 [14]: credited in part ("antecedents", p. 2); **attribution gap**
F6	*same result*	supporting	Orthonormal semicircular Wick polynomials (free Wick products) by signed deletion of disjoint adjacent pairs	Lemma 3.1 (p. 5) and its proof (p. 6); proof of Prop. 3.2 (p. 7); §1 (p. 2)	Effros–Popa 2003 [15], Kemp–Nourdin–Peccati–Speicher 2010 [16]: not cited; **attribution gap**
F7	*same result*	peripheral	Conjugate variables of a free semicircular perturbation as the projected rescaled noise	Lemma 3.1 (pp. 5–6); used on pp. 6, 8–9 and 11	Voiculescu 1998 [17]: cited for other content (definitions)

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1 About this document

The preprint. The preprint is *A finite-entropy separation of microstates and nonmicrostates free entropy*, dated September 25, 2026, with author line "OpenAI" [18]. It is one of the manuscripts in the `openai/math` release, produced by an internal OpenAI model. It constructs a bounded self-adjoint tuple X in a tracial von Neumann algebra with $-\infty < \chi(X) \leq \chi^*(X) - \frac{1}{2} < \infty$, where χ is Voiculescu's microstates free entropy and χ^* his nonmicrostates free entropy. Section 2 summarizes the paper and fixes the notation used throughout this document. The preprint cites one companion preprint from the same release, [19], as `FreeGroupFactors`, for a result that it contrasts with its own and does not use. It belongs to the same release, so it is not treated as earlier work here.

What this document reports. Each finding is a citation the preprint is missing: part of the preprint's mathematics (a technique, construction, estimate, proof route or result) that an identifiable earlier paper already contains, where the preprint does not cite that paper for it where it is used. "Earlier" means publicly available before the preprint's date. The one exception, overstated novelty, is described below. Not reported: textbook material in its textbook form; weaker versions of intermediate statements; and, for content the preprint uses, cases where it cites a later source that covers that content rather than the original. An earlier result on the same problem is reported even when later work that the preprint cites implies it. Findings are ordered by centrality, then by relationship in the order listed below, then by position in the preprint.

Relationships. A finding's relationship is the one between the preprint and the first earlier work listed for it (the principal), the work with the highest relationship in the order below; when several hold that relationship, however closely each matches, the one the others credit or build on, otherwise the earliest. Any other works listed relate to the same content more weakly or in part; the finding says how, including which work originated the idea when that is not the first.

same result. An earlier paper, or several earlier papers together, prove essentially the same statement.

same technique. The earlier technique, construction or theory is reused essentially intact, possibly in a new setting or with new ingredients around it.

adapted. The earlier technique's structure is followed, with a change at its core: a different object, or a key step replaced.

proof shortcut. A published result would replace or substantially shorten a large part of the argument.

same route. An earlier paper proposed the proof strategy, even heuristically, and the preprint carries it out.

fragment. One identifiable piece of the earlier technique is reused.

weaker result. An earlier paper proves a weaker, special-case or partial-range version of one of the preprint's main results, or of a statement it presents as a significant contribution.

Centrality. Each finding also gets a best-effort judgment of how much of the preprint's main argument depends on it: *main result* (reserved for earlier work that covers the main result itself, or most of it), *central*, *supporting* or *peripheral*. An earlier weaker result is never labelled main result. Each finding states the chain of results it feeds, so the basis for the judgment is visible.

Citations. The "Cited?" column of the summary names the earlier work and says what the preprint cites for the content in question: *not cited* (the work is not in the bibliography, and nothing the preprint cites credits it for this content), *cited for other content*, *credited elsewhere* (credited for this content, directly or through a cited location that credits it, but not where it is used, in a way that an expert reader who doesn't know the earlier work might not connect with it), or *credited in part*. A citation counts only when it is *for that specific content*; citing the same paper nearby for a different result does not count. Content the preprint credits where it is used, directly or through a cited location that itself credits the earlier work, is not reported, except when its novelty is overstated (below). For overstated novelty, the column says where the preprint does cite the work.

Flags.

Attribution gap. Marks the most serious missing citations: the relationship is same technique, adapted or same result; the centrality is main result, central or supporting; and the preprint cites the principal earlier work for this content nowhere.

Overstated novelty. The preprint does cite the earlier work for this content, but its abstract, a statement of its significant contributions, or the framing of a main result presents that content as new, explicitly or in the way a reader would reasonably take it. This is the one case in which content the preprint credits is reported.

How to check a claim. Locations in the preprint use its section, equation and theorem numbers and the page numbers printed in its PDF. Locations in other sources give the theorem, equation and printed page of the version listed in the bibliography. Each finding ends with a short checklist of the passages to put side by side.

Provenance. This document was generated by LLM agents (Claude Opus 5.5) running the precedence-review skill of jnshi/math-lineage (commit `e025345` for the section passes and commit `13ccc19` for the later passes), makes no claim of completeness, and has not been vetted by a human. It was run by Jonathan Shi, who also supplied the leads behind findings F1–F7, harvested from a run of an earlier version (v1) of the skill on this preprint, including his own lead behind finding F4. This run was also used to test updates to the skill.

2 The preprint in brief, and notation

Main result. Theorem 1.1 (p. 2): there exist $n \geq 2$ and a bounded self-adjoint tuple $X = (X_1, \ldots, X_n)$ in a von Neumann algebra with faithful normal tracial state τ such that

$$-\infty < \chi(X) \leq \chi^*(X) - \tfrac{1}{2} < \infty,$$

where χ is Voiculescu's microstates entropy with an operator-norm cutoff and a limsup over matrix sizes. The preprint presents this as a negative answer to the finite-entropy form of the equality question $\chi(X) > -\infty \Rightarrow \chi(X) = \chi^*(X)$ (§1, p. 1).

Entropies (§2, p. 3). $\Gamma_R(X; m, d, \varepsilon)$ is the set of self-adjoint $d \times d$ matrix tuples with coordinate norms at most R whose normalized traces of words of length at most m are within ε of those of X; $\chi_R(X) = \inf_{m,\varepsilon} \limsup_d [d^{-2} \log \mathrm{Vol}\, \Gamma_R(X; m, d, \varepsilon) + \frac{n}{2} \log d]$ and $\chi(X) = \sup_R \chi_R(X)$. For a tuple Z, the conjugate variables ξ_i satisfy $\tau(\xi_i P(Z)) = (\tau \otimes \tau)((\partial_i P)(Z))$ for noncommutative polynomials P, with ∂_i the free difference quotient; $\Phi^*(Z) = \sum_i \|\xi_i\|_2^2$, and, with S a free standard semicircular family,

$$\chi^*(Z) = \tfrac{n}{2} \log(2\pi e) + \tfrac{1}{2} \int_0^\infty \Big[\frac{n}{1+s} - \Phi^*(Z + \sqrt{s}\, S) \Big]\, ds \qquad (2), p.\ 3.$$

On the matrix side, $\mathrm{tr}_d = d^{-1} \mathrm{Tr}$, a standard GUE tuple $G^{(d)}$ has independent coordinates whose real Hilbert–Schmidt entries have variance $1/d$, and the normalized classical entropy of a random tuple with density f is $H_d = -d^{-2} \int f \log f + \frac{n}{2} \log d$.

The signal (§4, p. 7). Fix $p \geq 2$ and $n = p + 1$. $B_1, \ldots, B_p$ are free variance-one semicirculars, and Y, in a separate tensor factor, is uniform on the p equally spaced points $y_l = -1 + 2(l-1)/(p-1)$ of $[-1, 1]$, so Y commutes with every B_i. $C = (B_1, \ldots, B_p, Y)$, $c_i = \tau(C_i^2)$, $V_a = C + \sqrt{a}\, S$, $v_i(u) = u + c_i$, and $L(a) = \frac{1}{2} \sum_i \log(2\pi e(a + c_i))$ is the entropy of the semicircular family (or of the independent GUE law) with the same coordinate variances. $\kappa_m(C)$ is the order-m free-cumulant array of C. The tuple of Theorem 1.1 is $X = V_h$ for a small $h > 0$; the second noise scale is $t = p^{3/4}$, with p fixed large, as in (11), p. 9. The spectral projections of $X_n = Y + \sqrt{h}\, S_n$ for the intervals between the midpoints of the y_l are denoted e_l; see (12), p. 9.

The proof routes, as the preprint names them. The introduction says that the proof "compares two noise scales" (§1, p. 2).

- *The analytic estimate* (§§3–4): a squared-cumulant bound on the Fisher deficit $\Phi^*(V_u) - \sum_i v_i(u)^{-1}$ (Proposition 3.2, (6), p. 6), proved through the "Wick testing" identity of Lemma 3.1 (p. 5), together with the sparse cumulants of C (Lemma 4.1, p. 8), gives $\chi^*(V_a) > L(a) - \frac{1}{2}$ for large a, which is (9), p. 9. Finiteness, $\chi(X) > -\infty$, is Lemma 4.2 (p. 9).
- *The transfer between the scales*: an entropy-production lemma for GUE heat flow started from uniform laws $A^{(d)}$ on near-optimal microstate sets, Lemma 5.1 (p. 10): the inequality (13), $\limsup_d H_d(A^{(d)} + \sqrt{t}\, G^{(d)}) \geq \chi_R(X) + \frac{1}{2} \int_0^t \Phi^*(V_{h+s})\, ds$, which the preprint describes as using "the classical-to-free Fisher comparison of Jekel and Pi".
- *The matrix estimate* (§6): the evolved laws μ_d of $A^{(d)} + \sqrt{t}\, G^{(d)}$ retain a block structure, defined by projection decompositions $P = (P_l)$ of small rank and the compressions $\mathcal{D}_P D = (\sum_l P_l D_i P_l)_i$, with probability at least $\frac{1}{2}$, while the Gaussian reference law γ_d with variances $v_i(h + t)$ has it with probability at most e^{-4d^2} (Lemma 6.1, p. 12). Relative-entropy data processing turns this into a classical entropy deficit, and the proof of Theorem 1.1 combines it with Lemma 5.1 and (9)–(10) (pp. 13–14).

3 Findings

F1. Finite microstates entropy of a tuple perturbed by small free semicircular noise

Summary. Voiculescu proved that $\chi(X_1 + Y_1, \dots, X_m + Y_m) \geq \chi(Y_1, \dots, Y_m)$ when $Y_1, Y_2, \dots$ are free and of finite entropy and the X_j lie in $W^*(Y_{m+1}, Y_{m+2}, \dots)$ [1, Prop. 5.5, p. 431], and applied it to a tuple in a free group factor plus free semicircular noise (proof of Prop. 6.5, p. 435); the preprint cites this paper only for the definition of χ in §1 (p. 1). Lemma 4.2 (p. 9), the finiteness half of Theorem 1.1, proves $\chi(X) \geq \frac{n}{2}\log(2\pi h)$ for $X = C + \sqrt{h}\,S$, citing only the preprint's own convergence facts. *Same result*: as Y is atomic, $W^*(C) \cong L(\mathbb{F}_p) \bar{\otimes} \ell_p^\infty$ embeds trace-preservingly in $L(\mathbb{F}_2)$ by Voiculescu's compression theorem [2], not cited, and Prop. 5.5 then gives $\chi(X) \geq \frac{n}{2}\log(2\pi e h)$, larger by $n/2$. Jung's inequality $\chi(a+\varepsilon Z : Z) \geq \mathbb{P}_{2c\varepsilon\sqrt{n}}(a) + n\log\varepsilon + \underline{\chi}(Z)$, for Z free from a and $c = \max_i \|Z_i\|$ [3, proof of Lemma 2.2, p. 4], not cited, gives this bound directly for any a with matrix models, whose packing term $\mathbb{P}$ is then nonnegative (*same result*, later than Voiculescu's). Its proof is the lemma's argument, with noise uniform on microstate sets where the preprint takes GUE noise: microstates of the unperturbed tuple plus noise asymptotically free from them lie in the sum's microstate sets, so the noise alone bounds their volume and the tuple's own entropy, $-\infty$ here, plays no role; cf. Jekel's proof of [4, Claim 4.15, pp. 46–47], not cited, with the same structure for free semicirculars and a tensor-independent coordinate.

Correspondence.

	Preprint	Voiculescu [1]	Jung [3]
Tuple	C; models in dimensions pq_k (p. 8)	$X_j \in W^*(Y_{m+1}, \dots)$; here C, via $x \otimes \delta_l \mapsto E_{ll} \otimes x$ into $M_p(\mathbb{C}) \otimes L(\mathbb{F}_{1+p^2}) \cong L(\mathbb{F}_2)$ [12, (6.3), p. 172]	a with matrix models, so $\mathbb{P}_{2c\varepsilon\sqrt{n}}(a) \geq 0$
Noise	$\sqrt{h}\,S$, free from C, $0 < h < 1$	$Y_j = \varepsilon S_j$, free from the free group factor; $\varepsilon = \sqrt{h}$	εZ, Z free from a, $\varepsilon < 1/(2c+2)$; here $Z = S$, $c = 2$, $\varepsilon = \sqrt{h} < \frac{1}{6}$
Step	noised models in $\Gamma_{R_0}(X; m, d, \varepsilon)$ with probability $q_d \to 1$, density at most $(d/2\pi h)^{nd^2/2}$ (p. 9)	Props. 2.3, 3.8, 5.4 (pp. 414–430); no matrices	packing centres of $\Gamma_R(a)$ translated by noise microstates (m, γ)-free from them: disjoint sets in $\Gamma(a + \varepsilon Z : \varepsilon Z)$ (pp. 3–4)
Bound	$\chi(X) \geq \frac{n}{2}\log(2\pi h)$ (p. 9)	$\chi(X) \geq \chi(\sqrt{h}\,S) = \frac{n}{2}\log(2\pi e h)$	$\chi(X) \geq \chi(X : S) \geq \frac{n}{2}\log h + \underline{\chi}(S) = \frac{n}{2}\log(2\pi e h)$

How to check.

1. Lemma 4.2 (p. 9) against [1, Prop. 5.5, p. 431], whose printed conclusion has n where m is meant, and the display in the proof of Prop. 6.5 (p. 435) with $\varepsilon = \sqrt{h}$; the tensor description of C (§4, p. 7) against [12, (6.3), p. 172].
2. The proof of Lemma 4.2, with the models of p. 8, against the chain of inequalities in Jung's proof of [3, Lemma 2.2, pp. 3–4] and Jekel's proof of [4, Claim 4.15, p. 47].

F2. Microstates entropy bounded by the entropy of GUE-evolved microstate models minus the integrated free Fisher information

Summary. Jekel and Pi's proof of $\chi \leq \chi^*$ [5, proof of Thm 6.4] obtains, at the foot of p. 27 and for $\mathcal{B} = \mathbb{C}$, the inequality (13) of Lemma 5.1 (p. 10), keeping the entropies of the GUE-evolved models and bounding them by the Gaussian maximum only on p. 28. The preprint cites Jekel–Pi for an elementary proof of $\chi \leq \chi^*$ (p. 1) and, at the transfer, only for their Proposition 6.1, the classical-to-free Fisher comparison (pp. 2, 10); its overview of the proof of Theorem 1.1 (p. 2) presents the finite-time inequality that retains the terminal entropies as what Lemma 5.1 adds to that comparison. *same result*: Jekel–Pi prove (13) for the same ensembles, uniform laws on cutoff microstate sets, along an ultrafilter where the preprint selects one dimension per accuracy level, and the proof of Lemma 5.1 (pp. 10–11) follows theirs step by step, proving two inputs itself (Lemmas 2.1–2.2, pp. 3–5). Their approach compares χ with the integrated free Fisher information without large deviations: realize χ_R by uniform laws on near-optimal microstate sets (the realization of χ by random matrix models formulated in Jekel's appendix to Shlyakhtenko–Tao [6, Prop. B.7], which Jekel–Pi credit), add GUE noise, integrate the classical Fisher information by de Bruijn's identity, and bound it below in the limit by Φ^*, since matrix integration by parts turns the classical score tested against polynomials into the conjugate-variable functional. The preprint's own step comes next: its §6 bounds the retained entropies by a relative-entropy deficit (p. 2).

Correspondence.

Step	Preprint	Jekel–Pi [5], $\mathcal{B} = \mathbb{C}$
Models	Uniform on $\Gamma_R(X; k, d_k, 1/k)$, log-volume $\geq \chi_R(X) - 1/k$, one d_k per k; (14), p. 10	Uniform on $\Gamma_{\mathbf{R}}^{(n)}(\mathcal{O}_k)$, $h^{(n)} \geq \chi_{\mathbf{R}}^{\mathcal{U}} - 1/k$ on A_k; proof of Thm 4.10, pp. 18–19; Rem. 4.14, p. 19
Entropy gain	(15)–(16), p. 10: $\frac{d}{ds} H_d(A^{(d)} + \sqrt{s}\, G^{(d)}) = \frac{1}{2} I_d(s)$, I_d the normalized Fisher information	(3.2), p. 11: $h^{(n)}(\mathbf{X} + t^{1/2}\mathbf{S}) - h^{(n)}(\mathbf{X}) = \frac{1}{2}\int_0^t \mathcal{I}^{(n)}(\mathbf{X} + u^{1/2}\mathbf{S})\, du$
Fisher bound	$\liminf_d I_d(s) \geq \Phi^*(V_{h+s})$ via matrix integration by parts and Lemma 2.1; (17)–(18), p. 11	Prop. 6.1, p. 24: $\Phi^*(\mathbf{X} + u^{1/2}\mathbf{S}) \leq \lim_{n\to\mathcal{U}} \mathcal{I}^{(n)}(\mathbf{X}^{(n)} + u^{1/2}\mathbf{S}^{(n)})$; via Cor. 5.5 and Lemma 5.8, pp. 22–23
Result	(13): $\limsup_d H_d(A^{(d)} + \sqrt{t}\, G^{(d)}) \geq \chi_R(X) + \frac{1}{2}\int_0^t \Phi^*(V_{h+s})\, ds$	p. 27: $\lim_{n\to\mathcal{U}} h^{(n)}(\mathbf{X}^{(n)} + t^{1/2}\mathbf{S}^{(n)}) \geq \chi^{\mathcal{U}}(\mathbf{X}) + \frac{1}{2}\int_0^t \Phi^*(\mathbf{X} + u^{1/2}\mathbf{S})\, du$

How to check.

1. (13), p. 10 against the foot of [5, p. 27] with $\mathcal{B} = \mathbb{C}$ and the normalizations of p. 11; then p. 28 and the overview (p. 2). Lemma 4.9's proof (pp. 15–16) gives $\mathcal{U}$ with $\chi_{\mathbf{R}}^{\mathcal{U}} = \chi_{\mathbf{R}}$ at a cutoff above $\|X\|$; indices n_k in the sets A_k of Theorem 4.10's proof (p. 18), approaching the ultralimit, give (13) as stated.
2. The proof of Lemma 5.1 (pp. 10–11) step by step against those of Theorem 6.4 (p. 27) and Proposition 6.1 (pp. 25–26); Fatou replaces Corollary 3.9 (p. 9) and Lemma 6.3 (p. 26).

F3. A Fisher-deficit bound from higher-order Stein kernels lifted to semicircular chaos

Summary. Diez [7, Lemma 4, p. 29] bounds the free Fisher information of a semicircular smoothing, relative to a semicircular law, by the free Stein discrepancy of any order k, carrying to higher order the lemma of Fathi and Nelson [8, Lemma 2.4, p. 9], who introduced free Stein kernels and on whom he builds (p. 29); Cébron [9, Lemma 2.6, p. 5] extends their lemma, for one variable, to signals without regularity. The preprint cites none of these works, and its citations for the cumulant calculus and for conjugate variables do not cover this argument. Its proof of the squared-cumulant bound on the Fisher deficit, Proposition 3.2, (6) (pp. 6–7), with the lift (5) (p. 5) for $r \geq 2$, follows this argument; through (7)–(9) (pp. 8–9) it gives $\chi^*(V_{h+t}) > L(h+t) - \frac{1}{2}$ in the proof of Theorem 1.1 (p. 14). The shared approach, the free Stein-kernel argument (Fathi–Nelson; at higher order, Diez), writes the relative score as the conditional expectation of the signal's Stein kernel lifted into the semicircular chaos of the noise, an order-r kernel becoming an order-r Wick polynomial with weight $u^{-r/2}$, and bounds it by the chaos isometry. *same technique*: the kernels are the constant cumulant arrays $\kappa_{r+1}(C)$ of all orders $r \geq 2$ at once, and the lift is proved combinatorially, so, as for Cébron, no regularity of C is needed ($\Phi^*(C) = \infty$, as Y is atomic).

Correspondence. Preprint: $Z = V_u$, ξ_i its conjugate variables, $D_{\mathbf{j}}(P)$ the trace $\tau^{\otimes(r+1)}$ of the r-th iterated free difference quotient of P at Z. Diez: X is the preprint's C, $X(t) = e^{-t}X + \sqrt{1-e^{-2t}}\,\rho^{-1/2}S$ ((100), p. 28), E_t the conditional expectation onto $W^*(X(t))$, $\Xi(t)$ its conjugate variables, A_k an order-k Stein kernel; $Z = e^t X(t)$ for $\rho = 1$, $u = e^{2t} - 1$.

	Preprint	Diez [7]
Stein identity	$\tau(Z_i P(Z)) - v_i D_i(P) = \sum_{r\geq 2,\mathbf{j}} \kappa_{r+1}(C_i, C_{j_1}, \ldots, C_{j_r}) D_{\mathbf{j}}(P)$ (pp. 6–7)	$\langle \rho X, P(X)\rangle - (\tau\otimes\tau)\circ \mathrm{Tr}([\mathcal{J}P](X)^*) = \langle A_k, [\mathcal{J}^k P](X)\rangle_2$ (Def. 17, p. 18)
Lift	$D_{\mathbf{j}}(P) = u^{-r/2}\tau(U_{\mathbf{j}}(S)^* P(Z))$ (5), p. 5	order-k conjugate variables of $X(t)$: $\rho^{k/2}(1-e^{-2t})^{-k/2} E_t(U_k^{\mathbf{j}}(S))$ ((103), p. 28)
Residual	$Z_i - v_i\xi_i = E[R_i \mid W^*(Z)]$, $R_i = \sum_{r\geq 2,\mathbf{j}} u^{-r/2}\kappa_{r+1}(C_i, C_{j_1}, \ldots, C_{j_r}) U_{\mathbf{j}}(S)^*$ (p. 7)	$\Phi^*(X(t) \mid V_\rho) = c_t \langle \partial_S^{k,*}(-A_k), \Xi(t) - \rho X(t)\rangle_2$, $c_t = e^{-(k+1)t}\rho^{k/2}(1-e^{-2t})^{-k/2}$ ((123), p. 34)
Norm	$\sum_i \|R_i\|_2^2 = \sum_{r\geq 2} u^{-r}\|\kappa_{r+1}(C)\|_2^2$; contraction and $v_i \geq u$ give (6) (p. 7)	$\|\partial_S^{k,*}(A_k)\|_2 = \|A_k\|_2$ ((118), p. 31; (122), p. 34); Cauchy–Schwarz gives (106)

How to check.

1. $Z_i - v_i\xi_i = E[R_i \mid W^*(Z)]$ (p. 7) against [7, (107)–(110), pp. 29–30, and (123), p. 34], and against Cébron's first-order $\sqrt{1-e^{-2t}}\,\tau(F \mid F_t) = e^{-t}\tau(\partial_S^* A \mid F_t)$ [9, p. 6].
2. (5) against [7, (103), p. 28] under $Z = e^t X(t)$, which scales order-k conjugate variables by e^{-kt}; at $C = 0$, $u = 1$, (5) is Diez's definition $U_k^{\mathbf{j}}(S) = \partial_S^{k,*}(1^{\otimes(k+1)})$ (Def. 15, p. 17).
3. The proof of (6), p. 7, before $v_i \geq u$, against [7, (106), p. 29], applied formally (C meets neither Lemma 4's hypotheses nor $c_n = 1$): for $c_i = 1$ both bound $\Phi^*(V_u) - n/(1+u) = (1+u)^{-1}\Phi^*(X(t) \mid V_1)$, by $(1+u)^{-2}$ times $u^{-k}\Sigma_k^*(C \mid V_1)^2$ and $\sum_{r\geq 2} u^{-r}\|\kappa_{r+1}(C)\|_2^2$.

F4. A coordinate commuting with free semicirculars, kept through small free noise, forces an entropy deficit on matrix models

Summary. Jekel's counterexample in the setting of laws [4, §4.4, pp. 45–51] perturbs two free semicirculars and a semicircular tensor-independent of them by small free semicircular noise, obtaining a law μ with finite $\chi^{\mathcal{U}}$; bounded matrix models of a second, free law ν that are L^2-close to models of μ inherit the near-commutation and fall short of $\chi^{\mathcal{U}}(\nu)$ by an amount bounded away from zero. The preprint builds the tuple of its Theorem 1.1 on the same construction, in §4 (p. 7), and proves the same kind of deficit in Lemma 6.1 and the proof of Theorem 1.1 (pp. 12–14), but does not cite Jekel's paper anywhere; its one nearby citation bearing on this design is to Ge's argument for prime factors [14], given on p. 2 only as an antecedent for commutation constraints on microstate volumes in general. The shared approach follows Jekel's construction: small free noise makes χ finite, while the tensor coordinate's near-commutation survives and costs entropy to the matrix models that inherit it. *adapted*: the preprint makes the coordinate atomic, so that near-commutation becomes block structure, and puts the loss on the law's own evolved microstate ensembles μ_d, via a block event, a unitary-net covering and data processing rather than Jekel's conditioning and change of variables.

Correspondence.

	Preprint	Jekel [4]
Construction	$X = C + \sqrt{h}\,S$, h small, Y atomic in a separate tensor factor (§4, p. 7); $\chi(X) > -\infty$ (Lemma 4.2, p. 9)	$\mathbf{X} = (1-\varepsilon)^{1/2}(S_1, S_2, S_3) + \varepsilon^{1/2}(S_1', S_2', S_3')$: S_1, S_2 free, S_3 tensor independent of them (pp. 45–46); $\chi^{\mathcal{U}}(\mu) > -\infty$ (Claim 4.15, pp. 46–47)
Kept after noise	(12), p. 9: spectral projections e_l of X_n near the $e_l(0)$, which commute with all B_i: $\tau(e_l) < 3/(2p)$, $\sum_{i,l}\|e_l X_i e_l\|_2^2 > 3p/4$	$\|[X_j, X_3]\| \le 24\varepsilon^{1/2}$, $j = 1, 2$ (p. 46)
In matrix models	(19), p. 12: the midpoint spectral partition P of $A_n^{(d)}$ has rank $P_l \le 2d/p$, $\sum_{i\le p,\,l}\|P_l A_i^{(d)} P_l\|_{2,d}^2 > p/2$	$\lim_{n\to\mathcal{U}}\|[X_1^{(n)}, X_3^{(n)}]\|_{L^2} \le 24\varepsilon^{1/2}$, passed to $Y_j^{(n)}$ at cost $2R\|Y_j^{(n)} - X_j^{(n)}\|_{L^2}$ (p. 48)
Entropy deficit	$\limsup_d H_d(A^{(d)} + \sqrt{t}\,G^{(d)}) \le L(h+t) - 1$ (p. 14), via a block event, μ_d-probability $\ge 1/2$, γ_d-probability $\le e^{-4d^2}$ (Lemma 6.1, p. 12)	for R-bounded models $\mathbf{Y}^{(n)}$ of ν and $\mathbf{X}^{(n)}$ of μ: $e^{a/2}[(25+6R)\varepsilon^{1/2} + 2Rb^{1/2}] \ge \varepsilon^{1/4}$, $a = \chi^{\mathcal{U}}(\nu) - \lim_{n\to\mathcal{U}} h^{(n)}(\mathbf{Y}^{(n)})$, $b = \lim_{n\to\mathcal{U}}\|\mathbf{X}^{(n)} - \mathbf{Y}^{(n)}\|_{L^2}^2 - d_{W,\mathrm{CEP}}(\mu,\nu)^2$ (Claim 4.16, (4.5), p. 47)

How to check.

1. §4, p. 7 and (12), p. 9 against Jekel's construction and commutator bound [4, pp. 45–46].
2. (19), p. 12 against the passage of the commutator bound to $\mathbf{X}^{(n)}$ and $\mathbf{Y}^{(n)}$ [4, p. 48].
3. Lemma 6.1 and p. 14 against Claim 4.16 [4, pp. 47–51]: the preprint's deficit falls on the law's own ensembles, Jekel's on models of ν (p. 45), by different estimates.

F5. Bounding the measure of matrix tuples with a common block structure by a unitary net over fixed blocks

Summary. Voiculescu proved in the third paper of his free-entropy series [10] (FE III below) that finite free entropy excludes property Γ and Cartan subalgebras, by a covering argument expounded by Biane [11, Lemme 6.3, pp. 289–290] and by Mingo and Speicher [12, §7.7.1, Lemma 9, pp. 196–198]; Dykema's many-block version [13, Lemma 2.2, pp. 3–7] is the closest in form. The preprint cites none of these works: its §6 cites nothing, and its introduction (§1, p. 2) says only that controlling microstate volumes by commutation and projection geometry has "antecedents" in Ge's argument for prime factors [14], and cites Szarek [20] for the net bound. The Gaussian half of Lemma 6.1 (p. 12), proved on p. 13, adapts FE III's approach to show that under γ_d the first p coordinates rarely put a fixed fraction of their energy into the blocks of some decomposition into p projections of rank at most $2d/p$: conjugate the decomposition to fixed blocks, replace the unitary by a point of an operator-norm net paid for once, and bound the measure at each net point, where the gain accumulates over the coordinates. The insight, that one common unitary makes the constraint strong (Mingo and Speicher, p. 198), is Voiculescu's; his theorem [12, Theorem 7(i), p. 195] already gives $\chi(C) = -\infty$, as Y is central in $W^*(C)$. *adapted*: the reduction is followed step by step, but the estimate at each net point becomes a chi-square tail under γ_d, not a volume of balls with a thin off-diagonal factor as $\omega \to 0$, and the event ranges over all rank lists.

Correspondence. Preprint: $D_{[p]} = (D_1, \dots, D_p)$, $\|D\|_{2,d}^2 = \sum_i \operatorname{tr}_d(D_i^* D_i)$, $v = v_i(h+t)$ for $i \le p$, and P^0 the standard coordinate blocks of a rank list. Biane: d variables, a microstate $(M_1, \dots, M_d)$ of size N, and a projection Π of rank $q = [N\tau(P)]$ with $\|[M_i, \Pi]\|_2 < 2\omega$.

	Preprint, proof of Lemma 6.1	FE III, after Biane
Fixed blocks	each decomposition is a conjugate UP^0U^*; compression $\mathcal{D}_{P^0}$ (p. 13)	$\Pi = U \operatorname{diag}(I_q, 0)U^*$; off-diagonal blocks C_i of U^*M_iU with $\frac{2}{N}\operatorname{Tr}(C_iC_i^*) \le 4\omega^2$ (p. 289)
Net	operator-norm δ-net of $U(d)$, $(1+2/\delta)^{2d^2}$ points, $\delta = 1/(64\sqrt{v})$ (p. 13)	$(C\delta^{-1})^{N^2-q^2-(N-q)^2}$ points U_s of a Szarek net of $U(N)/(U(q)\times U(N-q))$ (p. 290)
Conjugation loss	$\|U^*D_{[p]}U - V^*D_{[p]}V\|_{2,d} \le 2\delta\|D_{[p]}\|_{2,d}$, at most $\sqrt{p}/8$ on the event (p. 13)	commutator at most $2\omega + 8\delta$ at the net point (p. 290)
Per net point	$\gamma_d(\|\mathcal{D}_{P^0}(U^*D_{[p]}U)\|_{2,d}^2 \ge p/64) \le e^{d^2(\log 2 - p/(256v))}$ by chi-square on $p\sum_l \operatorname{rank}(P_l^0)^2 \le 2d^2$ of pd^2 dimensions; union over $(d+1)^p$ rank lists (p. 13)	volume of $\big(U_s^*(B(\sqrt{4N}) \times B((2\omega + 8\delta)\sqrt{N}) \times B(\sqrt{4N}))U_s\big)^d$ over $s \in S$, with $\delta = \omega$, giving $\chi \le C_1 + C_2 \log\omega$ (p. 290)

How to check.

1. The Gaussian bound's proof (p. 13) against the proof of Lemme 6.3 [11, pp. 289–290] and [12, pp. 196–198]: in both, the net is paid for once, against a gain from all p, respectively $d \ge 2$, coordinates.
2. The introduction's citations of Ge and Szarek (§1, p. 2) against [12, p. 195], which credits Theorem 7(i)–(ii) to Voiculescu and (iii), primeness, to Ge.

F6. Orthonormal semicircular Wick polynomials given by deleting adjacent pairs

Summary. Effros and Popa gave an explicit formula for the q-Wick product; at $q = 0$ it is a sum over gap-free diagrams, that is, sets of disjoint adjacent pairs [15, (12), p. 12]. Kemp, Nourdin, Peccati and Speicher (KNPS), following Biane and Speicher [21], state that $Y \mapsto Y\Omega$ is an isometry into the full Fock space sending products of Chebyshev polynomials over runs of equal indices to orthonormal elementary tensors [16, Prop. 3.1 and (3.2), p. 29]. The preprint cites neither: it names "orthogonality of semicircular Wick polynomials" as a tool (§1, p. 2), citing Speicher and Nica–Speicher there only for the cumulant calculus, defines $U_{\mathbf{j}}(S)$ by deleting disjoint adjacent pairs (proof of Lemma 3.1, p. 6), proves its orthonormality (Lemma 3.1, p. 5) by a cumulant cancellation, and uses it in the proof of Proposition 3.2 (p. 7). *same result*: the definition is Effros and Popa's formula at $q = 0$ term by term; orthonormality, which they do not state, follows at once from their characterization $u\Omega = f_1 \otimes \cdots \otimes f_n$ of the Wick product (p. 7), defined following Bożejko, Kümmerer and Speicher [22]; and since pairs that straddle two runs carry $\delta = 0$, $U_{\mathbf{j}}$ is the KNPS product over runs, which on a single run is the classical expansion of U_m.

Correspondence.

	Preprint	Effros–Popa (EP) [15]; KNPS [16]
Family	$U_{\mathbf{j}}(S)$, $\mathbf{j} = (j_1, \ldots, j_r)$: sum over sets F of disjoint adjacent pairs of the word $S_{j_1} \cdots S_{j_r}$ with the pairs in F deleted, each with factor $-\delta_{j_a j_{a+1}}$ (proof of Lemma 3.1, p. 6)	$:\xi_1 \cdots \xi_r: = \sum_{\gamma \in \mathsf{GF}(r)} (-1)^{\#(\gamma)} v(\gamma)$, $v(\gamma)$ the covariances of the pairs times the word of the singletons (EP (12), p. 12; v, p. 5); $\xi_a = \varphi(e_{j_a})$ gives $\mathbb{E}(\xi_a \xi_{a+1}) = \delta_{j_a j_{a+1}}$
Runs	runs $k_1^{m_1}, \ldots, k_\ell^{m_\ell}$: $U_{\mathbf{j}}(S) = U_{m_1}(S_{k_1}) \cdots U_{m_\ell}(S_{k_\ell})$ (not stated); e.g. $U_{(1,1,2)} = S_1^2 S_2 - S_2 = U_2(S_1) U_1(S_2)$	$U_{n_1}(X(h_{k_1})) \cdots U_{n_r}(X(h_{k_r}))$, $k_i \neq k_{i+1}$ (their r counts runs; KNPS (3.2), p. 29), U_n as in (2.11), p. 28
Orthonormality	$\tau(U_{\mathbf{j}}^* U_{\mathbf{k}}) = \delta_{\mathbf{jk}}$ (Lemma 3.1, p. 5; proof at $C = 0$, $u = 1$, p. 6)	$u\Omega = f_1 \otimes \cdots \otimes f_r$ (EP p. 7), so $\mathbb{E}(u^* v) = \langle v\Omega, u\Omega \rangle$; $Y \mapsto Y\Omega$ isometric (KNPS Prop. 3.1, p. 29)

How to check.

1. The first paragraph of the proof of Lemma 3.1 (p. 6) against [15, (12) and (13), p. 12] with $\xi_a = S_{j_a}$: a diagram is gap free when every pair is $\{a, a+1\}$ (gap g, p. 4; GF, p. 11).
2. Orthonormality in Lemma 3.1 (p. 5) and its proof (p. 6) against $u\Omega = f_1 \otimes \cdots \otimes f_n$ [15, p. 7] at $q = 0$ with orthonormal f_a.
3. Factor $U_{\mathbf{j}}$ over runs and compare with [16, Prop. 3.1 and (3.2), p. 29]. On one run of length m the deletion rule gives $\sum_i (-1)^i \binom{m-i}{i} x^{m-2i}$, the classical expansion of $U_m(x)$.

F7. Conjugate variables of a free semicircular perturbation as the projected rescaled noise

Summary. Voiculescu's fifth paper on free entropy and Fisher information (FE V) shows that, for a free semicircular family S free from X, the conjugate variables of $X + \sqrt{\varepsilon}\, S$ always exist

and equal $\varepsilon^{-1/2}E[s_i \mid W^*(X+\sqrt{\varepsilon}\,S)]$ [17, Cor. 3.9]; Major and Shlyakhtenko [23, §2.1, p. 4] and Cébron [9, p. 6] cite it under that number, and Fathi and Nelson cite its Ornstein–Uhlenbeck form as FE V Prop. 3.7 [8, p. 7]. The preprint cites FE V only in its introduction, for the definitions of conjugate variables, free Fisher information and χ^*, and states the formula without citation as the last statement of Lemma 3.1 (p. 5), proved as the $r=1$ case of its Wick-testing identity (p. 6). *same result*: in the preprint's notation ($X=C$, $\varepsilon=u$, $Z=C+\sqrt{u}\,S$) the statement is Voiculescu's formula, whose content is that the rescaled noise $u^{-1/2}S_i$ already satisfies the conjugate relations of Z, so that its projection onto $L^2(W^*(Z))$ is the conjugate variable for every C; the preprint checks these relations with noncrossing cumulants. The formula supplies the conjugate variables in the proof of Proposition 3.2 (p. 6), where it yields $\Phi^*(V_u)\le n/u$, which makes the integrals in (8) and (13) finite (pp. 8–9 and 11); its existence statement lets Lemma 2.1 give (18) in the proof of Lemma 5.1 (p. 11). Both uses lead to Theorem 1.1.

Correspondence.

	Preprint	FE V, as quoted
Setting	C a centered bounded self-adjoint n-tuple, S a standard semicircular family free from C, $Z=C+\sqrt{u}\,S$, $u>0$ (Lemma 3.1, p. 5)	$X=(x_1,\dots,x_d)$, $N\subseteq M$ a unital $*$-subalgebra, $S=(s_1,\dots,s_d)$ a free $(0,1)$-semicircular family, free from $N\cup X$; $\varepsilon>0$ [23, p. 4]
Formula	$\xi_i=u^{-1/2}E[S_i \mid W^*(Z)]$ (p. 5)	$\mathcal{J}(x_i+\sqrt{\varepsilon}\,s_i : N)=\varepsilon^{-1/2}E_\varepsilon(s_i)$, with E_ε the conditional expectation onto $W^*(N, X+\sqrt{\varepsilon}\,S)$ [23, p. 4]
Existence	the conjugate variables of Z exist, for every such C (p. 5)	for any self-adjoint $x_1,\dots,x_n$ and $x_j(t)=e^{-t}x_j+\sqrt{1-e^{-2t}}\,\rho^{-1/2}s_j$, the conjugate variables of $X(t)$ always exist and $\xi_j(t)=\sqrt{\rho}\,(1-e^{-2t})^{-1/2}\mathcal{E}_t(s_j)$ [8, p. 7]

How to check.

1. The last display of Lemma 3.1 (p. 5) against [23, §2.1, p. 4] with $N=\mathbb{C}$, $X=C$ and $\varepsilon=u$: then $\mathcal{J}(x_i+\sqrt{\varepsilon}\,s_i:\mathbb{C})$ is the i-th conjugate variable ξ_i of the tuple Z, and $E_\varepsilon=E[\,\cdot\mid W^*(Z)]$.
2. The same display against [8, p. 7]: $x_j(t)=e^{-t}(x_j+\sqrt{\varepsilon}\,s_j)$ with $\varepsilon=(e^{2t}-1)/\rho$, and conjugate variables scale as $\xi_j(\lambda Z)=\lambda^{-1}\xi_j(Z)$; with $\lambda=e^{-t}$ their formula becomes $\xi_j(X+\sqrt{\varepsilon}\,S)=\varepsilon^{-1/2}E[s_j \mid W^*(X+\sqrt{\varepsilon}\,S)]$.
3. The last sentence of the proof of Lemma 3.1 (p. 6): for $r=1$, $U_i(S)=S_i$ and $D_i(P)=(\tau\otimes\tau)(\partial_i P(Z))$, so (5) says $\tau(u^{-1/2}S_i\,P(Z))=(\tau\otimes\tau)(\partial_i P(Z))$, that is, the rescaled noise satisfies the conjugate relations of Z; projecting it onto $L^2(W^*(Z))$ gives the right-hand side in item 1.

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